Initial Models Report

Beyond Accuracy: A Clinical Validation Approach to Cardiovascular Disease Prediction Using Machine Learning

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DSE6311 Capstone

This project evaluates machine learning model performance through a clinical validation lens rather than standard accuracy metrics. To assess clinically appropriate risk, how should model performance be evaluated, given the trade-off between sensitivity and specificity?
Model optimization and validation will balance the delicate trade-off between false negatives and false positives. The final model may not be the most accurate based solely on accuracy/ROC-AUC metrics.
A tuned ensemble method, such as Random Forest or XGBoost optimized for sensitivity, will outperform Logistic Regression on recall and ROC-AUC. The optimal model will be defined as the sensitivity-specificity combination that best aligns with clinical risk thresholds and patient outcomes.

Methods:

- Box – Tidwell test performed to assess the linearity of the log-odds assumption
- Age was binned into clinical categories using scaled threshold equivalents to address the identified violation
- VIF confirmed no perfect multicollinearity (performed in Preprocessing and Feature Engineering)
- Independence satisfied by data design, one patient per row
- Encoding note: In response to feedback regarding the variable thal, the variable was confirmed to be treated as nominal rather than ordinal. One-hot encoding via pd.get_dummies with drop_first = True, produced two binary indicator columns – thal_6.0(Fixed Defect), and thal_7.0 (Reversible defect), with thal_3.0 (Normal) as the reference category. The numeric values 3, 6, and 7 were not treated as ordinal.

The following applies to LR Baseline, LR SMOTE, and LR CSL initial models

- Hyperparameters c = 1.0, L2 penalty, random_state = 42, max_iter = 1000.
- Stratified k-fold cross-validation (k = 10) applied to the training set
- Train and test scores returned to assess for overfitting
- Evaluated test set is the standard model.
- Consistent evaluation methodology ensures differences reflect model performance and not the evaluation approach.
- CSL models will have a class_weight = ‘balanced’

Results:

- Box – Tidwell initial results: age violated the linearity assumption ($p = 0.001$), trestbps ($p = 0.172$), thalach ($p = 0.848$), chol_boxcox ($p = 0.078$) all passed.
- Following age binning, post-remediation on Box – Tidwell confirmed that all remaining continuous variables satisfy linearity trestbps ($p = 0.226$), thalach ($p = 0.785$), chol_boxcox ($p = 0.178$)
- All three Logistic Regression assumptions confirmed met

LR – Training Set

- Cross validation ROC-AUC: 0.887 +- 0.066
- Recall (sensitivity): 0.744 +- 0.154, below clinical target of >= 85-90%
- Train/test gaps are all below 6%, no concerning overfitting detected
- Age assumption violation did not produce severe overfitting, will continue using binned scaled threshold equivalents.

LR – Test Set

- Sensitivity: 74.4% - below clinical target, 10 CVD cases missed
- Specificity: 91.1% - exceeds clinical target
- ROC – AUC: 0.932
- 4 false positives – unnecessary follow-ups (see Figure 10)
- No overfitting detected – test performance consistent with CV results
- Explore threshold adjustment in later models (default 0.5)

ROC Curve Interpretation (see Figure 11)

- At FPR ~0.2, the curve crosses the sensitivity target of 85%, reaching ~90% sensitivity.
- Corresponding specificity at this ppoint is 80%, within clinical target range.
- Flat section from FPR 0.20 – 0.30 indicates no additional sensitivity gain beyond this threshold
- Optimal threshold likely sits at the point where curve crosses 85% target at FPR 0.20
- Further threshold analysis will pinpoint optimal cutoff value.

Coefficients Inspection

- Reversible thalassemia defect (thal_7.0) was the strongest predictor (coefficient 1.463) – consistent with its known role in CVD.
- Asymptomatic chest pain (cp_4) was the second strongest predictor
- Being male (1.018) and number of blocked vessels (ca 0.822) were also strong positive predictors.

- Maximum heart rate (thalach – 0.360) and non-anginal chest pain (cp_3 -0.689) were the strongest negative predictors – higher heart rate associated with absence of CVD, consistent with EDA findings.
- St_severity (0.410) showed a positive relationship with CVD – validating the feature engineering performs in Week 4.
- Age_binned showed a small coefficient 0.070 – consistent with a weak age signal and the linearity violation identified during assumption testing.
- Fbs showed minimal predictive contribution (-0.276) – consistent with EDA findings of weak class separation.

**** Limitations – Proxy Leakage*
*Several top predictors including thal, ca, exang, restecg, and st_severity are diagnostic findings measured during the cardiovascular workup rather than pre-existing risk factors. This represents a form of proxy leakage, meaning that the model is partly learning from findings discovered during the diagnostic process rather than from independent predictors. This does not invalidate the model but scopes it as a diagnostic classification tool for patients already undergoing cardiovascular workup rather than a prospective population-level risk prediction tool. This limitation is consistent with how the UCI Cleveland dataset has been used in published literature and will be acknowledged in the final deliverable. ****

LR - SMOTE

LR – SMOTE Training set

- SMOTE model outperformed standard model across all cross-validation metrics
- Recall improved by 8.3 percentage points (0.744 → 0.827)
- ROC-AUC improved 0.887 → 0.902
- No concerns of overfitting
- SMOTE model shows smaller train – test gaps (1-3%) vs standard model (4-6%) suggesting better generalization.
- SMOTE confirmed as beneficial for this dataset

LR – SMOTE Test Set

- Sensitivity improved from 74.4% to 79.5%, 2 fewer missed cases, false negatives reduced from 10 to 8
- Specificity remains unchanged at 91.1%
- Accuracy improved 83.3% to 85.7%
- ROC-AUC improved marginally from 0.9323 to 0.936
- Sensitivity still below >= 85% target, threshold adjustment and CSL required

LR – SMOTE Confusion Matrix (see figure 13)

- True positives increased from 29 to 31
- False negatives reduced from 10 to 8
- False positives unchanged at 4
- True negatives unchanged at 41

ROC Comparison (see figure 14)

- ROC curves for standard and SMOTE nearly identical, AUC difference of 0.004
- SMOTE curve sits slightly above standard in upper left region
- Both models cross 85% threshold at ~FPR 0.20
- SMOTE benefited primarily from sensitivity improvement rather than overall discriminative ability.

LR - CSL

LR – CSL Training set

- CSL model showed minimal improvement over standard LR model. Recall 0.754 compared to 0.744.
- SMOTE outperformed CSL on all metrics.
- No overfitting detected.
- ROC-AUC nearly identical 0.889 vs 0.887.

LR – CSL Test Set

- CSL model produced identical test results to SMOTE – accuracy 85.7%, sensitivity 79.5%, specificity 91.1%.
- Compared to standard model, both SMOTE and CSL independently improved sensitivity from 74.4% to 79.5%.
- Both methods addressed mild class imbalance. Combined SMOTE + CSL may produce further improvement.

LR – CSL Confusion Matrix (see figure 15)

- Identical results to SMOTE

ROC Comparison

- Will do final comparison of all four curves after CSL + SMOTE model.

LR – CSL + SMOTE

LR – CSL + SMOTE Training set

- CSL + SMOTE produced identical CV results to SMOTE alone across all metrics.
- SMOTE created 50/50 balance, applying CSL created redundancy.
- CSL independently matches SMOTE performance, combining provides no additional benefit.

LR – CSL + SMOTE Test Set

- All three models produce identical results, standard model is the only difference.
- SMOTE will be preferred approach going forward as it's simple and effective.

LR – CSL + SMOTE Confusion Matrix

- Skipped due to redundancy of output.

ROC Comparison (All Models, see figure 16)

- Besides standard model, all curves nearly identical.
- Standard LR is slightly below other models in upper left region.
- All four models outperform random classifier.
- All four curves cross 85% sensitivity target at ~0.20 FPR.

Discussion:

Box – Tidwell assumption testing identified a linearity violation for age. This was addressed by age binning using scaled threshold equivalents, and creating a categorical variable instead. The logistic Regression baseline performed well with ROC-AUC of 0.932 and specificity of 91.1%, but sensitivity of 74.4% fell below the clinical target of >= 85%. SMOTE improved sensitivity 74.4% → 79.5% and caught two additional CVD cases, without increasing false positives. CSL independently matched SMOTE performance, but CSL + SMOTE provided no additional benefits. All three variants of the standard Logistic Regression model provided identical results. The sensitivity target of >= 85% has still not been achieved. ROC curve analysis suggests target could be achievable through threshold adjustment at ~0.20 FPR.

Upon reflection and response to feedback regarding SMOTE, approaches to class imbalance have been reconsidered. Random oversampling was considered, in an effort to have true and non-synthetic data creation but was rejected due to risk of overfitting from exact record duplication. Cost-sensitive learning requires class imbalance to apply meaningful penalty

differences and uses real patient data. On the original training set, the minority CVD class received a higher weight (1.087 vs 0..926) causing the model to penalize missed CVD cases ~17% more heavily. On the SMOTE training set, both classes received equal weights of 1.0, and no additional penalty was applied. CSL was redundant on an already balanced dataset.
CSL addresses imbalance AND embeds clinical risk simultaneously.

These findings can connect back to the research question, suggesting that standard accuracy metrics alone do not capture the appropriate sensitivity and specificity tradeoffs and require optimization.

Next Steps:

- Random Forest and XGBoost models will be built with and without CSL on original training set as primary imbalance handling strategy.
- Hyperparameter tuning for both ensemble methods.
- Threshold analysis across all models to identify clinically optimal cutoff.
- Full model comparison across all variants.
- SHAP analysis for feature importance interpretation.
- Decision Curve Analysis planned for Week 7 to assess clinical utility.

Appendix

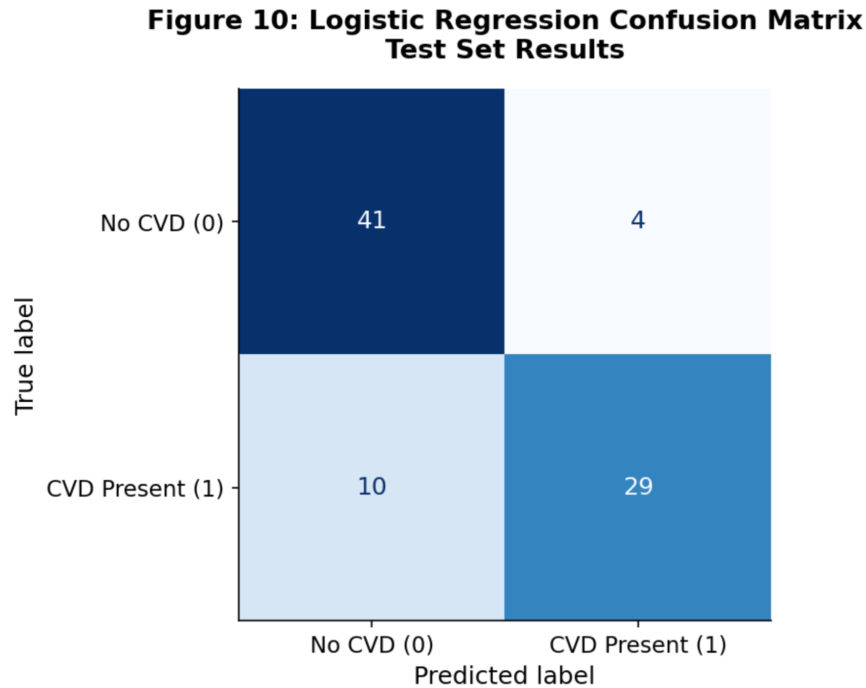

Figure 10: Confusion Matrix Logistic Regression Baseline Standard Model.

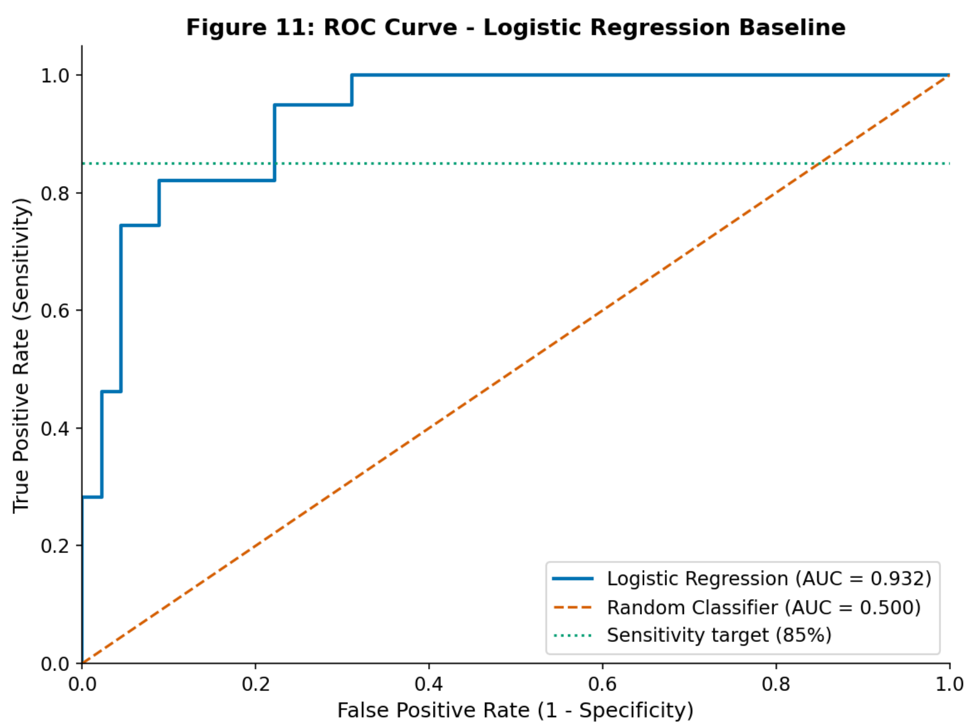

Figure 11: ROC Curve Logistic Regression Standard Baseline

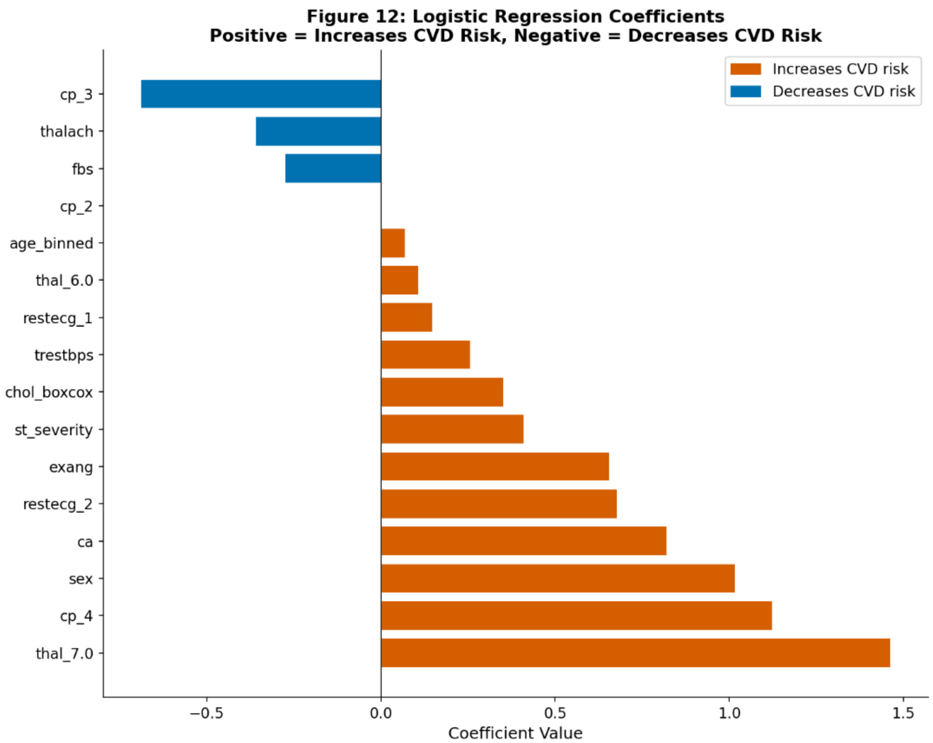

Figure 12: LR coefficient plot.

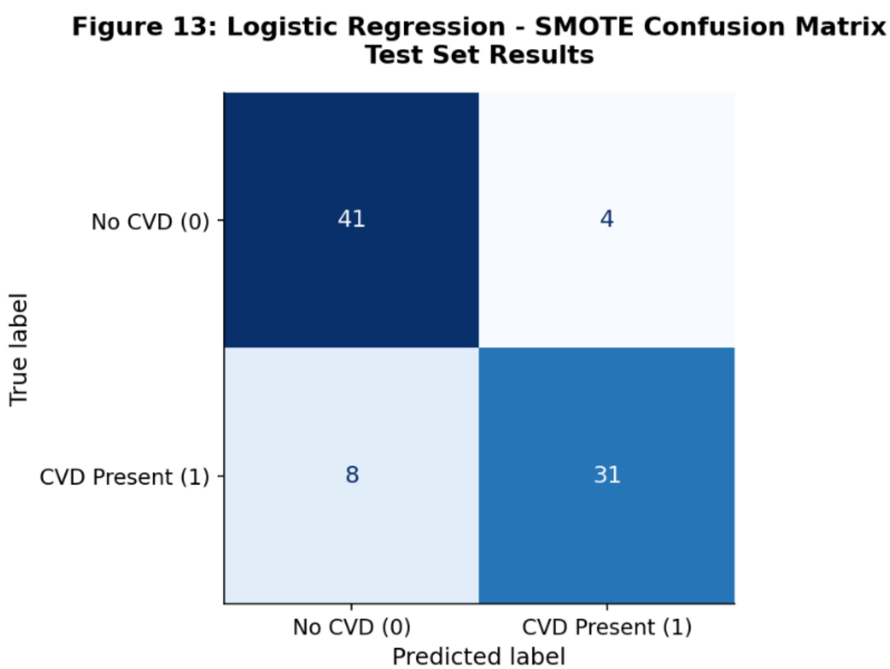

Figure 13: LR – SMOTE Confusion Matrix

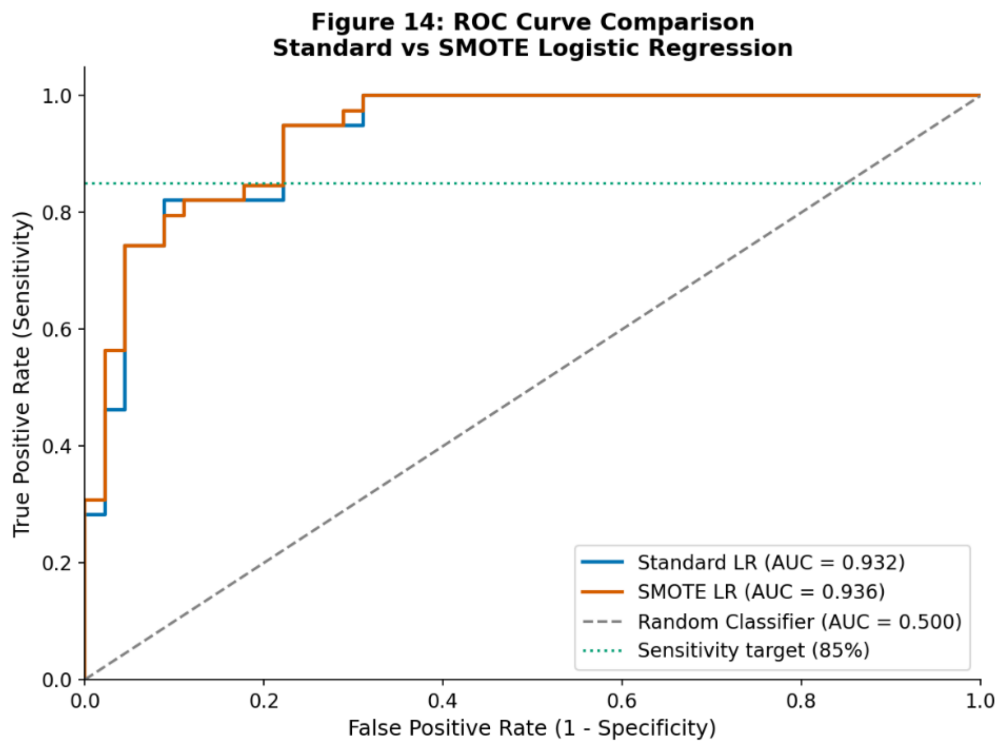

Figure 14: ROC Curve comparison of LR Standard vs SMOTE

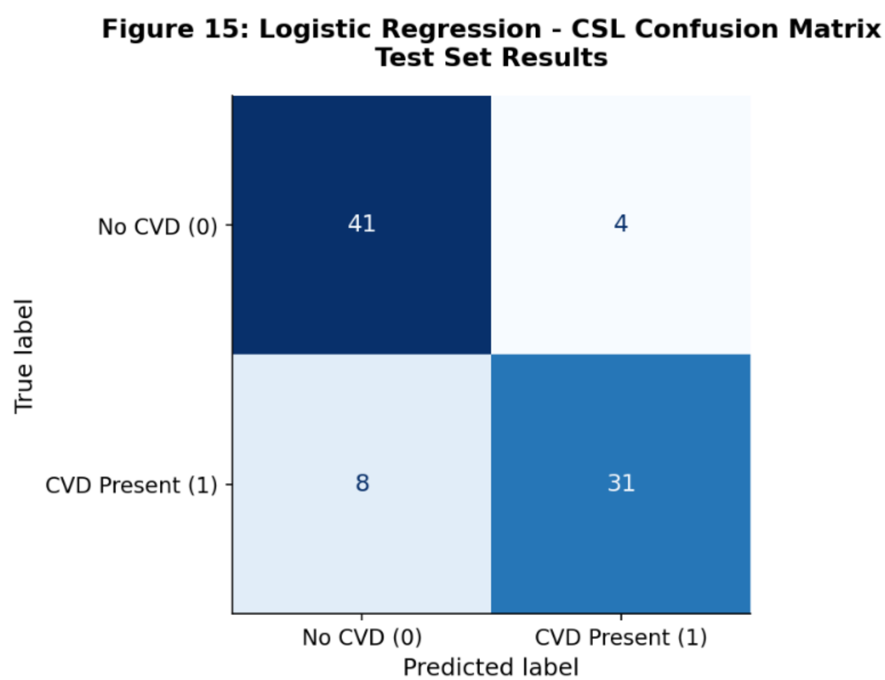

Figure 15: Confusion Matrix for LR CSL

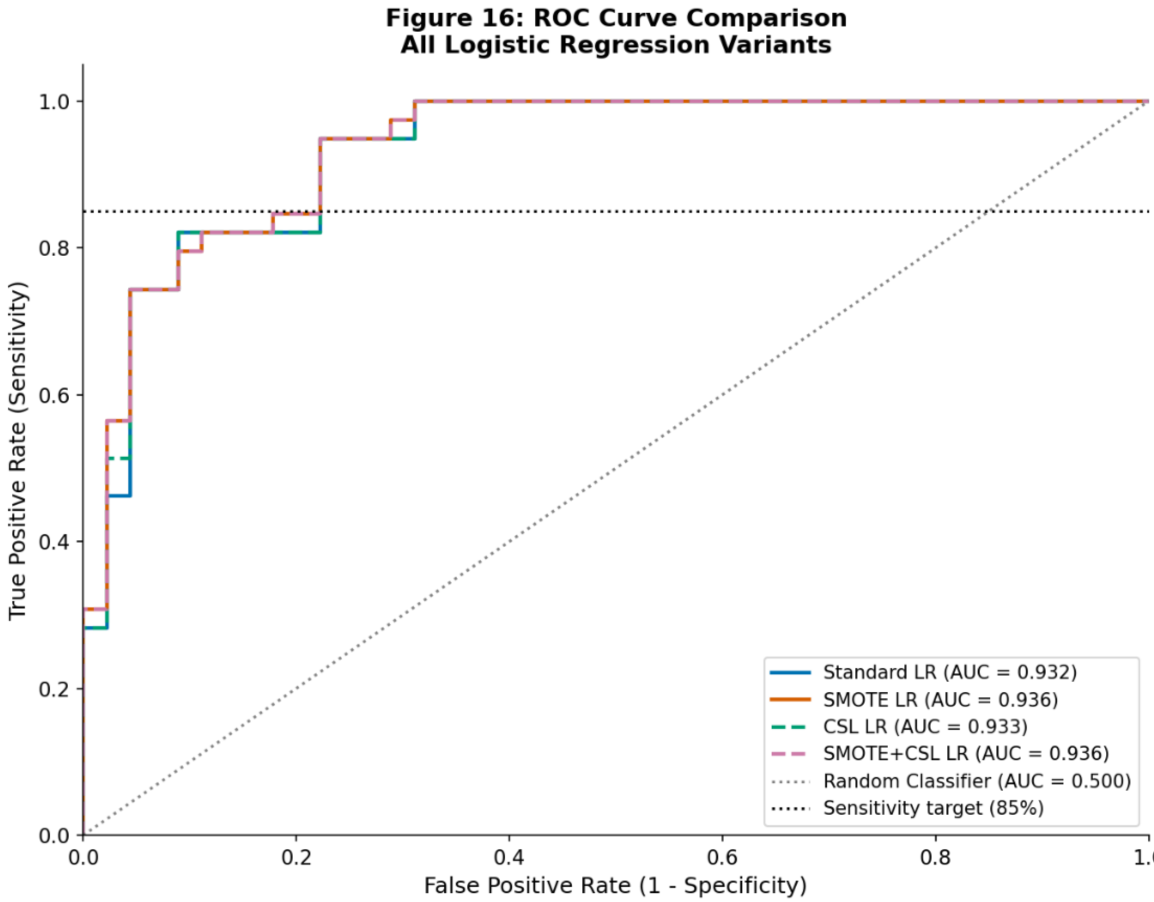

Figure 16: ROC Curve Comparison of all four model variants

Table 3: Logistic Regression Model Comparison - Test Set Results

	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	FN	FP
Standard LR	0.833	0.879	0.744	0.911	0.806	0.932	10.0	4.0
SMOTE LR	0.857	0.886	0.795	0.911	0.838	0.936	8.0	4.0
CSL LR	0.857	0.886	0.795	0.911	0.838	0.933	8.0	4.0
SMOTE+CSL LR	0.857	0.886	0.795	0.911	0.838	0.936	8.0	4.0

Table 3: Test set results comparison of all four model variants.

Data Dictionary:

Variable	Type	Description	Values/Range
age_binned	Categorical	Age group by categories to address BT violation	0=Young (<45), 1=Middle (45-60), 2=Older (>60)
sex	Categorical	Biological sex	0 = Female, 1 = Male
cp	Categorical	Chest pain type	1 = Typical Angina, 2 = Atypical Angina, 3 = Non-Anginal Pain, 4 = Asymptomatic
trestbps	Continuous	Resting blood pressure on admission (mmHg)	94 - 200, note : standardized via StandardScaler during preprocessing
chol_boxcox	Continuous	Box - Cox transformed cholesterol serum	standardized | Original range: 126-564 mg/dl, transformed and standardized via StandardScaler
fbs	Categorical	Fasting blood sugar > 120 mg/dl	0 = Normal, 1 = Elevated
restecg	Categorical	Resting electrocardiographic results	0 = Normal, 1 = ST-T Abnormality, 2 = LV Hypertrophy
thalach	Continuous	Maximum heart rate achieved during exercise (bpm)	71 - 202, note: standardized via StandardScaler during preprocessing
exang	Categorical	Exercise induced angina	0 = No, 1 = Yes
st_severity	Categorical	ST depression and slope of peak exercise severity index	0 = Normal, 1 = Equivocal, 2 = Positive, 3 = Strongly Positive
ca	Categorical	Number of major vessels colored by fluoroscopy	0 - 3
thal_	Categorical	Thalassemia type	_3.0 = refrence category, _6.0 = Fixed Defect, _7.0 = Reversible Defect
target	Binary Target	Cardiovascular disease diagnosis	0 = No CVD, 1 = CVD Present